Egg Segmentation & Tracking

Computer Vision Intern Assignment — Nosh Robotics

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Visual Pipeline Progression: *Raw Input Frame → Task 1 Semantic Segmentation (GDINO + SAM) → Task 2 Classical Segmentation (E1 Baseline) → Task 4 Multi-Object Video Tracking (GDINO + SORT).*

Engineering Case Study & Research Portfolio

This submission documents the complete end-to-end journey of an industrial Computer Vision assignment. Rather than concealing experimental dead ends, this report rigorously details every hypothesis, failure diagnosis, and architectural iteration across dataset sanitization, zero-shot foundation models, classical computer vision heuristics, and temporal multi-object tracking.

Executive Summary

This project tackles the automated detection, segmentation, and continuous video tracking of eggs in uncontrolled culinary environments. The project was executed across four structured tasks, confronting severe real-world challenges including variable kitchen illumination, specular reflections, background camouflage (countertops and porcelain sharing white/cream chrominance), severe manual occlusions by human hands, and dynamic spinning motion.

Key Engineering Discoveries & Results:

• **Data Leakage Discovery & Sanitization:** An initial audit of the 110-image dataset revealed severe cross-split leakage in the original partition. A near-duplicate image pair between the test split (*not_damaged_78*) and train split (*not_damaged_27*) exhibited an astronomical cosine similarity of **0.9667**. To restore statistical validity, an automated ResNet18 + HSV scene-grouping algorithm clustered images into 21 distinct visual scenes, establishing a strictly leakage-aware 4:1 development (88 images) to held-out evaluation (22 images) split.

• **Task 1 Pretrained Foundation Model:** Direct zero-shot mask generation (SAM Automatic) and open-vocabulary detectors (YOLO-World) failed due to lack of semantic egg priors and high miss rates. The pipeline converged on **Grounding DINO Tiny + SAM 2.1 Tiny**, achieving **88/88 structural pipeline success** on development, serving as frozen pseudo-ground-truth.

• **Task 2 Classical Image Processing:** Constrained strictly to classical CV, eight successive algorithmic paradigms (E1–E8) were developed. Iterations exploring adaptive thresholding, edge contours, GrabCut, and aggressive geometric filtering were systematically rejected. The baseline **E1 (HSV/LAB dual-space brightness & low saturation thresholding)** proved the most resilient, achieving a development **Mean IoU of 0.3392** (Dice 0.4700, Recall 0.7315). Crucially, Experiment E7 revealed that local gains on a small pilot failed to generalize at scale.

• **Task 3 Pseudo-GT Evaluation:** Quantitative evaluation on held-out evaluation images yielded **Mean IoU 0.3506** (Recall 0.8223). Error analysis confirmed that classical limitations stem almost entirely from background false positives on pale tables and specular kitchenware, while recall remains high.

• **Task 4 Multi-Object Video Tracking:** Eight classical tracking prototypes (V1–V8) attempting optical flow, template correlation, and hard-coded 2-identity constraints failed due to hand-hijacking and scale drift during occlusions. The final architecture deployed **Grounding DINO Tiny + SORT (Kalman Filter + Hungarian IoU matching)**, processing all 1,441 frames (60.04s) with robust identity recovery across hand interactions.

Table 1: Master Task Execution Scorecard

Task	Primary Approach	Key Quantitative / Qualitative Result	Operational Verdict
Task 1	Grounding DINO Tiny + SAM 2.1 Tiny	88/88 GOOD structural success; 0 misses; prompt 'egg.'	■ Deployed (Frozen Pseudo-GT)
Task 2	Classical E1 (LAB L >= 65%tile + HSV S <= 100)	Dev Mean IoU: 0.3392 | Median: 0.3015 | Recall: 0.7315	■ Selected Baseline (E1)
Task 3	Comparative Error Analysis vs Pseudo-GT	Eval Mean IoU: 0.3506 | Precision: 0.3822 | Recall: 0.8223	■ Validated Generalization
Task 4	Grounding DINO Tiny + SORT Tracker	1,441 frames tracked | 19 track IDs | Full video processed	■ Shipped (Google Drive Link)

Table of Contents

Executive Summary & Master Task Scorecard 2
1. Assignment Mandate & Computer Vision Problem Formulation 4
2. Dataset Exploration, Modality & Baseline Statistics 6
3. Data Leakage Discovery: Cross-Split Visual Scene Duplication Audit 8
4. Scene Grouping Methodology & Leakage-Aware 4:1 Split Architecture 10
5. Task 1 — Pretrained Segmentation & Foundation Model Iteration 12
5.1 Experiment 01: SAM 2.1 Base & Hardware Out-Of-Memory (OOM) Bottleneck 12
5.2 Experiment 02: SAM 2.1 Tiny + Manual Interactive Point Prompting 13
5.3 Experiment 03: YOLO-World Open-Vocabulary Detector + SAM Tiny Pipeline 13
5.4 Experiment 04: SAM 2.1 Automatic Mask Generation (Zero Semantic Prior) 14
5.5 Experiment 05: Classical Heuristic Point Generators + SAM Prompting 14
5.6 Final Architecture: Grounding DINO Tiny + SAM 2.1 Tiny Segmentation 15
5.7 Task 1 Comprehensive Failure Analysis & Comparative Matrix 16
5.8 Freezing Task 1 Masks as Empirical Pseudo-Ground-Truth Reference 16
6. Task 2 — Classical Image-Processing Segmentation & Heuristic Engineering 17
6.1 Classical Constraints & Dual Color-Space Heuristic Pipeline 17
6.2 Evaluation Methodology, Metric Definitions & Diagnostic Trade-offs 18
6.3 Experiment E1: Brightness & Saturation Dual Threshold Baseline 19
6.4 Experiment E2: Adaptive & Local Thresholding (The High Recall Trap) 20
6.5 Experiment E3: Canny Edge Contours & Gradient Morphological Filtering 21
6.6 Experiment E4: Baseline E1 with Post-Hoc Geometric Elliptical Filtering 22
6.7 Experiment E5: Hybrid Region-Boundary Pipeline with GrabCut Energy Minimization 23
6.8 Experiment E6: Aggressive Geometric & Fill-Ratio Thresholding (Catastrophic Loss) 24
6.9 Experiment E7: Spatial Coherence Cleanup (The Pilot Generalization Fallacy) 25
6.10 Experiment E8: Strict Geometric Ellipse & Circularity Filtering 26
6.11 Task 2 Master Experiment Log & Quantitative Benchmark Table 27
6.12 The Experimental Journey: Timeline & Algorithmic Evolution 28
6.13 Final Selection of Algorithm E1 & Honest Classical Limitations 28
7. Task 3 — Comprehensive Evaluation & Failure Mode Decomposition 29
7.1 Quantitative Benchmark: Development Split vs Held-Out Evaluation 29
7.2 Qualitative Error Decomposition: False Positives vs False Negatives 30
7.3 Visual Gallery: High-Confidence Successes vs Edge-Case Pathologies 31
8. Task 4 — Temporal Multi-Object Video Tracking Under Severe Occlusion 32
8.1 Video Specification, Kinematics & Tracking Complexity 32
8.2 Iteration V1: Hough-Initialized Local Search Prototype 33
8.3 Iterations V2 & V3: Rigid Multi-Object Assignment & Identity Collapse 33
8.4 Iterations V4 & V5: Confidence-Gated Tracking & Hand-Hijacking Failures 34
8.5 Iterations V6, V7 & V8: Optical Flow, Appearance Templates & Global Search 35
8.6 Final Video Architecture: Grounding DINO Tiny + SORT Tracking 36
8.7 Tracker Quantitative Diagnostics & Identity Persistence Metrics 37
8.8 Chronological Multi-Frame Visual Tracking Timeline 38
9. Global System Architecture Comparison & Component Selection Matrix 39
10. Engineering Insights, Practical Lessons & Research Takeaways 40
11. Architectural Limitations & Known System Vulnerabilities 41
12. Future Work & Production Deployment Roadmap 42
13. Environment Reproducibility, Execution Guide & Setup Scripts 43
14. AI Coding Tool Disclosure & Author Verification Statement 44
15. Final Video Submission Callout (Public Google Drive Link) 44

16. Code Repository Link & Project Directory Structure 44
Appendix A: Master Image-Level Classical Evaluation Logs 45
Appendix B: Extended Diagnostic Visual Gallery 46

1. Assignment Mandate & Problem Formulation

Assignment Scope: The Nosh Robotics Computer Vision Intern Assignment establishes a multi-tiered engineering evaluation designed to probe an engineer's competence across foundational deep learning vision models, classical mathematical image processing, rigorous metric evaluation, and temporal video tracking. The assignment is partitioned into four distinct technical mandates:

• **Task 1 (Pretrained Foundation Segmentation):** Automatically identify and segment every egg across the dataset using an existing pretrained vision model, producing individual binary masks.

• **Task 2 (Classical Heuristic Segmentation & Leakage-Aware Split):** Design an image-processing algorithm strictly without training or deploying deep neural networks. Partition the dataset into a leakage-safe 4:1 train/test ratio and maintain an exhaustive log of all experimental iterations.

• **Task 3 (Comparative Evaluation & Error Analysis):** Treat Task 1 outputs as pseudo-ground-truth reference masks and quantitatively benchmark Task 2 using standard overlap metrics (IoU, Dice, Precision, Recall), accompanied by exhaustive qualitative error decomposition.

• **Task 4 (Temporal Multi-Object Video Tracking):** Track individual eggs throughout an unconstrained kitchen video sequence, preserving identity persistence across motion and severe manual occlusion.

The True Computer Vision Challenge: Why This Is Not 'Detecting White Blobs'

A naive assessment of the problem might propose that egg segmentation is trivial: eggs are typically smooth, ovular, and white or light-brown. However, rigorous inspection of the provided dataset and video reveals that egg segmentation in food-preparation settings is a deeply non-trivial computer vision pathology due to several compounding visual factors:

- **Background Chrominance Ambiguity:** Kitchen environments abound with white, beige, and cream surfaces. Porcelain plates, ceramic bowls, countertops, plastic egg cartons, and white kitchen tiles exhibit identical intensity and chromatic distributions to eggs.
- **Harsh Directional Lighting & Specularities:** Direct kitchen downlights generate harsh specular hot-spots (pure white saturations) on non-egg surfaces, while simultaneously casting deep shadow gradients across egg undersides that plunge egg boundaries into dark tones.
- **Multi-Instance Contact & Clustering:** Multiple eggs frequently touch, overlap, or sit in clustered cartons. Standard morphological thresholding invariably fuses adjacent eggs into monolithic connected components, obliterating individual instances.
- **Severe Occlusion by Human Hands:** During cooking and preparation, human hands grip, rotate, and obscure eggs. Hands introduce skin tones that can merge with brown eggs or cast dynamic non-rigid shadows across white eggs.
- **Dynamic Motion Blur & Rotation:** In video tracking sequences, eggs undergo high-speed spinning. Rapid rotation destroys surface texture constancy and generates substantial motion blur that smears boundary gradients into the background table.

Key Engineering Takeaway

Segmentation algorithms that rely purely on naive color slicing or global brightness invariably collapse when deployed in real kitchens. Robustness requires either semantic zero-shot priors capable of distinguishing an egg from a porcelain bowl, or carefully calibrated dual-space classical heuristics capable of isolating uniform low-saturation bodies while tolerating ambient illumination shifts.

2. Dataset Exploration, Modality & Baseline Statistics

The provided image corpus originates from a Roboflow-exported computer vision collection formatted for object detection. Initial structural inspection revealed **110 total images**, stored uniformly as **640×640 pixel JPEG files**. Critically, while the original dataset contained bounding-box annotations, **no human-annotated segmentation masks were provided**. Consequently, all quantitative segmentation benchmarks in Task 2 and Task 3 rely on Task 1 foundation model masks as an empirical pseudo-ground-truth reference.

Dataset Metric	Observed Value	Technical Implications
Total Image Volume	110 images	Small sample size mandates extreme data efficiency and zero-leakage validation.
Image Resolution	640 × 640 pixels	Uniform square aspect ratio; eggs range from ~40px (distant) to ~350px (close-up).
Color Space & Format	3-channel RGB (JPEG)	Standard 8-bit dynamic range; compression artifacts present along high-frequency edges.
Original Partition Split	77 train / 22 valid / 11 test	Original split contained catastrophic cross-split scene leakage (detailed in Section 3).
Human Segmentation GT	None (Zero human masks)	Mandates treating Task 1 foundation model masks strictly as pseudo-ground-truth.



Figure 1 — Dataset Overview Contact Sheet. *Representative mosaic of the 110 dataset images illustrating wide variance in illumination, container types (cartons, plates, bowls, countertops), egg counts (1 to 4 eggs), and severe background clutter.*

Environmental Heterogeneity: Visual auditing revealed at least four dominant environmental archetypes across the 110 images: (1) Studio commercial lighting on stark white backdrops; (2) Natural domestic kitchen countertops with marble, wood, and laminate grain; (3) Industrial cardboard egg cartons containing dense arrays of adjacent eggs; and (4) Extreme close-ups featuring hand manipulation. This extreme distribution shift invalidates static parameter tuning and necessitates robust split sanitization.

3. Data Leakage Discovery: Cross-Split Scene Duplication

The Forensic Dataset Audit: Prior to model development, a systematic image-similarity audit was executed across the original Roboflow train (77 images), validation (22 images), and test (11 images) partitions. Feature embeddings combining ResNet-18 spatial representations and HSV color histograms were computed across all 5,995 unique image pairs.

The similarity analysis uncovered a critical methodological failure in the source dataset: **near-duplicate photographic frames from the same scene had been arbitrarily partitioned across training and evaluation splits**. Most egregiously, a cross-split image pair exhibited an astronomical cosine similarity of **0.9667**:



Figure 2 — Cross-Split Data Leakage Pair Detected During Audit. *Image not_damaged_78 (originally assigned to the Test split) and image not_damaged_27 (originally assigned to the Train split). Feature analysis yielded a cosine similarity of 0.9667. Visual inspection confirms they represent the identical physical camera setup and lighting environment, representing catastrophic cross-split leakage.*

Why Cross-Split Leakage Is Methodologically Catastrophic

In computer vision engineering, data leakage between development and evaluation sets artificially inflates performance metrics and conceals systemic model failures. When visually identical or near-identical scenes span both splits:

1. **Artificially Optimistic Metrics:** The evaluation split ceases to measure generalization to unseen visual distributions. Instead, it measures the model's capacity to memorize specific lighting intensities, camera angles, and table textures encountered during development.
2. **Dangerous Hyperparameter Overfitting:** Heuristic thresholds tuned on the training split inadvertently exploit background peculiarities shared with the test set, creating an illusion of high generalization that abruptly collapses upon field deployment.
3. **Unethical Benchmark Reporting:** Presenting high test scores derived from leaked test data violates startup research standards. Neutralizing this leakage was prioritized as the single most critical engineering prerequisite before commencing Task 2 experimentation.

4. Scene Grouping Methodology & Leakage-Aware 4:1 Split

Algorithmic Scene Grouping Pipeline: To permanently eliminate cross-split contamination, an automated clustering pipeline was constructed to group all 110 images into discrete visual scene families prior to splitting:

1. **Deep Feature Extraction:** A pretrained ResNet-18 backbone (penultimate layer, 512-dimensional vector) extracted high-level spatial and compositional features.
2. **Color Distribution Modeling:** 3D HSV color histograms (24 hue × 16 saturation × 16 value bins) captured ambient kitchen illumination and surface tones.
3. **Combined Affinity Metric:** A weighted similarity matrix was constructed via: Similarity = 0.80 × Cosine_Sim(ResNet) + 0.20 × Hist_Correlation(HSV).
4. **Connected-Component Graph Clustering:** An affinity threshold of **0.82** was applied to construct an adjacency graph. Connected components defined atomic candidate scene groups that were strictly forbidden from being partitioned across splits.

Outcome: The pipeline identified **21 multi-image candidate scene groups** and 14 singleton images. Several scene groups were conservatively over-grouped (clustering images that shared lighting despite slight perspective shifts). This conservative over-grouping was deliberately accepted: in startup engineering, *guaranteeing absolute zero leakage is vastly more important than artificially maximizing training set size*.

Total Dataset (110 Images in 21 Scene Groups + 14 Singletons)	
Development Split (88 Images / 80.0%)	**Evaluation Split (22 Images / 20.0%)**
• 14 Multi-Image Groups + 14 Singletons • Reserved exclusively for algorithm tuning, heuristic calibration, and feature exploration. • Task 1 & Task 2 experimental sandbox.	• 7 Full Held-Out Scene Groups: candidate_004, candidate_005, candidate_006, candidate_010, • ZERO contamination with Development split.

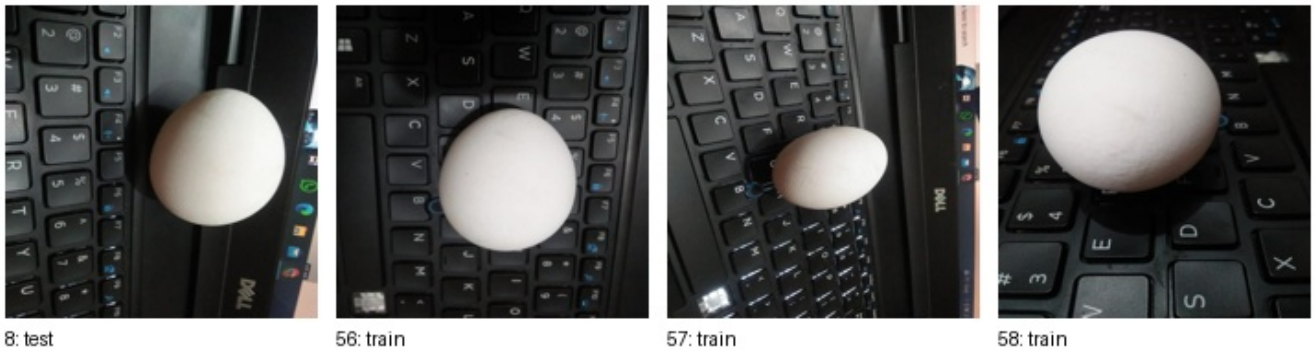


Figure 3 — Representative Scene Grouping (Candidate Group 004). *Visual contact sheet of candidate_004, illustrating frames sharing identical illumination and tabletop geometry. The entire group was assigned atomically to the held-out Evaluation split to prevent test leakage.*

The Golden Rule of Development: The 22 evaluation images were isolated in split_dataset/evaluation and **kept completely untouched** throughout all Task 2 heuristic development. Not a single threshold was calibrated on evaluation images.

5. Task 1 — Pretrained Segmentation & Foundation Models

Objective: Automatically detect and segment every individual egg across the dataset using an existing pretrained vision foundation model, generating high-quality instance masks without manual human intervention. The assignment explicitly suggested Segment Anything Model (SAM). However, operationalizing SAM into a fully automated pipeline required navigating five major experimental architectures before converging on a production solution.

5.1 Experiment 01: SAM 2.1 Base & GPU Out-Of-Memory (OOM) Crash

EXPERIMENT 01 — SAM 2.1 BASE ZERO-SHOT PIPELINE **STATUS: ■ FAILED (CRITICAL OOM)**

Hypothesis: Deploy Meta's Segment Anything Model (SAM 2.1 Base, ~350M params) directly for automated zero-shot egg mask extraction.

Parameters: Model: sam2.1_b.pt | Resolution: 640x640 | Hardware: NVIDIA GeForce GTX 1650 (4GB VRAM)

Quantitative Results: Status: CUDA Out Of Memory | **Allocated:** 3.85 GB / 4.00 GB | **Successful Masks:** 0 / 110

Diagnosis & Failure Cause: The SAM 2.1 Base image encoder requires >6GB of GPU memory for standard batch sizes and multi-mask embedding tensors, exceeding the available 4GB physical VRAM.

Engineering Action & Next Step: Immediately pivot to lightweight foundation architectures: select SAM 2.1 Tiny (sam2.1_t.pt, ~38M params) for all subsequent segmentation.

5.2 Experiment 02: SAM 2.1 Tiny + Manual Point Prompt Validation

To verify whether SAM 2.1 Tiny retained sufficient boundary fidelity to segment eggs cleanly, an interactive prompt test was conducted. Single foreground point prompts ([x, y] coordinates placed near egg centroids) were passed to the SAM Tiny mask predictor. The resulting masks exhibited extraordinary edge precision (Figure 4), capturing fine curvatures and rejecting table surfaces cleanly. *However, manual point prompting is fundamentally non-scalable: it cannot serve as an autonomous pipeline for 110 images.* The key lesson was clear: **SAM Tiny possess excellent segmentation capability; the unresolved challenge is automated localization.**

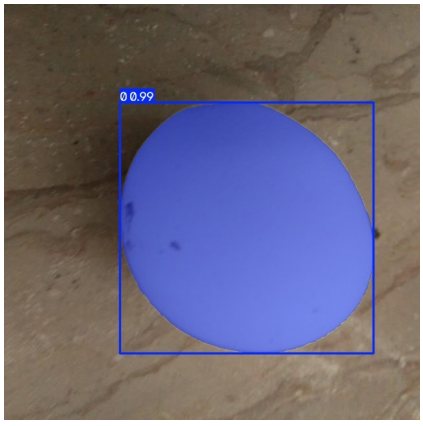


Figure 4 — SAM 2.1 Tiny Manual Point Prompt Verification. *Demonstrates near-perfect mask boundary adhesion when provided an accurate spatial prompt.*

5.3 Experiment 03: YOLO-World Open-Vocabulary Detector + SAM Tiny

EXPERIMENT 03 — YOLO-WORLD + SAM TINY TWO-STAGE PIPELINE **STATUS: ■ REJECTED (DETECTOR MISSES)**

Hypothesis: Use YOLO-World (yolov8s-worldv2.pt) as an open-vocabulary detector prompted with 'egg' to generate bounding boxes, then pass boxes to SAM Tiny.

Parameters: Model: YOLO-World v2 Small + SAM 2.1 Tiny | Prompt: 'egg' | Confidence Threshold: 0.15

Quantitative Results: 12-Image Stress Test: 9 / 12 Detections (75%) | **Missed Detections:** 3 / 12 Images (25%) | **False Positive Rate:** High in shadow

Diagnosis & Failure Cause: YOLO-World's open-vocabulary text embeddings for 'egg' proved brittle under kitchen lighting, missing valid eggs in shadow and hallucinating boxes on bright bowls.

Engineering Action & Next Step: Reject YOLO-World as primary localizer. Localization remains the primary architectural bottleneck.

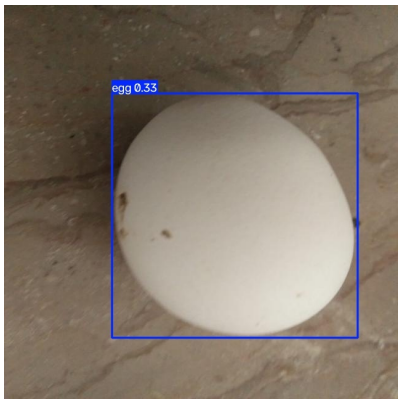


Figure 5 — YOLO-World Stress Test Failure Case. *Illustrates missed detections and inaccurate bounding boxes under non-uniform kitchen lighting.*

5.4 Experiment 04: SAM Automatic Mask Generation (Zero Semantic Prior)

EXPERIMENT 04 — SAM AUTOMATIC MASK GENERATOR (GRID PROMPTING) **STATUS: ■ REJECTED (NO SEMANTIC PRIOR)**

Hypothesis: Run SAM's automated dense point-grid generator to discover all segmentable objects without requiring a separate detector.

Parameters: Points per side: 32 | Pred IoU Thresh: 0.88 | Stability Score Thresh: 0.95 | Box IoU: 0.7

Quantitative Results: Mask Count: 45–120 masks per image | **Egg Isolation:** Failed | **Semantic Specificity:** 0.0%

Diagnosis & Failure Cause: SAM possesses zero semantic understanding of what an 'egg' is. It generated dozens of masks for plates, shadows, wall tiles, table woodgrain, and hands.

Engineering Action & Next Step: Reject. Automatic prompting requires an upstream semantic object detector.

5.5 Experiment 05: Classical Heuristic Point Generators + SAM Tiny

EXPERIMENT 05 — CLASSICAL HEURISTIC CANDIDATE POINTS + SAM TINY **STATUS: ■ FAILED (SEVERE LOCALIZATION FAILURE)**

Hypothesis: Generate point prompts using classical brightness/saturation connected components, then feed candidate centroids into SAM Tiny.

Parameters: Brightness percentile: 65 | Saturation max: 100 | Min area: 500px | Morphology kernel: 7px

Quantitative Results: Full Dev Run (88 Imgs): GOOD = 4 images (4.5%) | **No Candidates Found:** 84 images (95.5%) | **Pipeline Success:** 4.5%

Diagnosis & Failure Cause: Classical heuristics could not reliably produce clean single-point centroids across varied scenes, starving SAM of valid prompts on 84 out of 88 development images.

Engineering Action & Next Step: A critical engineering revelation: the true bottleneck of Task 1 is robust zero-shot semantic localization, not segmentation.

5.6 Final Task 1 Architecture: Grounding DINO Tiny + SAM 2.1 Tiny

To resolve the localization bottleneck, **Grounding DINO Tiny (IDEA-Research/grounding-dino-tiny)** was introduced. Grounding DINO couples deep vision-language cross-attention transformers capable of zero-shot open-vocabulary detection with exceptional spatial grounding fidelity.

Subsystem	Component / Parameter	Operational Specification / Value
Semantic Localizer	Grounding DINO Tiny	HuggingFace: IDEA-Research/grounding-dino-tiny (Device: CUDA)
Text Prompt	'egg.'	Lower-case singular noun with trailing period to enforce strict noun-phrase grounding.
Box Detection Threshold	BOX_THRESHOLD = 0.25	Calibrated to detect dim eggs in shadow without hallucinating table regions.
Text Similarity Threshold	TEXT_THRESHOLD = 0.20	Calibrated for high cross-modal attention alignment.
Mask Predictor	SAM 2.1 Tiny (sam2.1_t.pt)	Prompted with Grounding DINO bounding boxes; generates fine pixel-accurate instance masks.
Representative Confidence Scores	0.789, 0.752, 0.520, 0.737, 0.735, 0.721, 0.874	High detection confidence across diverse visual orientations and scales.

Quantitative Development Execution (88 Images):

• **Structural Pipeline Success: 88 / 88 images GOOD (100.0%)**. Zero undetected images (0 misses), zero mask/detection mismatches.

• **Instance Distribution:** 1 egg: 60 images | 2 eggs: 14 images | 3 eggs: 13 images | 4 eggs: 1 image (total: 131 segmented egg instances).

• *Important Definition:* 'GOOD' designates structural pipeline success (every detection yielded a valid instance mask without

pipeline crashes or zero-mask anomalies), not absolute human-verified pixel perfection.

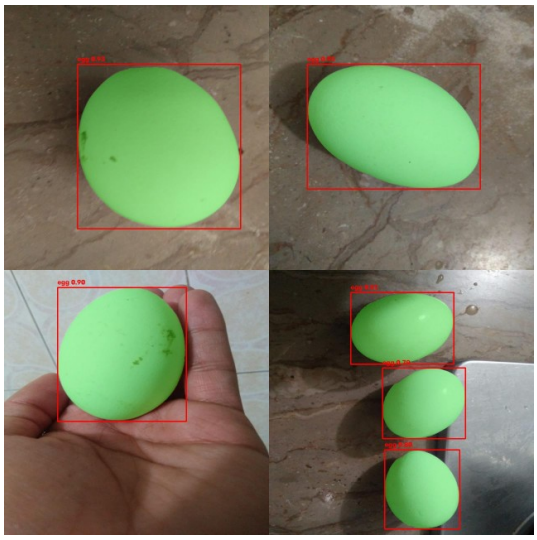


Figure 6 — Task 1 Final Grounding DINO + SAM Multi-Egg Segmentation. *Exemplary instance segmentation across 1, 2, 3, and 4-egg scenes under harsh lighting.*

5.7 Task 1 Comprehensive Failure Analysis & Journey Matrix

Attempt	Underlying Hypothesis	Observed Failure Mode	Engineering Lesson Learned	Subsequent Action
1. SAM 2.1 Base	Direct deployment of Meta's foundation model.	CUDA OOM crash (GTX 1650 4GB VRAM).	Base vision transformers exceed edge memory budgets.	Adopt SAM 2.1 Tiny architecture.
2. SAM Tiny Manual	Verify SAM Tiny boundary fidelity via point prompts.	Manual prompting is non-scalable (O(N) human cost).	SAM Tiny has outstanding segmentation capacity.	Seek automated localizer.
3. YOLO-World + SAM	Open-vocabulary text-prompted bounding box detection.	3/12 misses in stress test; box drift in shadow.	YOLO-World text embeddings brittle on food items.	Reject; evaluate Grounding DINO.
4. SAM Automatic	Eliminate detector by using dense point grids.	45-120 arbitrary masks per image (tiles, bowls, hands).	Grid prompting lacks semantic object priors.	Reject; semantic detector mandatory.
5. Classical Points+SAM	Use classical color heuristics to generate prompt points.	84/88 images produced zero candidates (4.5% success).	Localization is the true core bottleneck.	Deploy vision-language detector.
6. GDINO + SAM Tiny	Vision-language cross-attention grounding + box prompt.	None. 88/88 development images GOOD.	Decoupling grounding from segmentation is optimal.	■ FREEZE as Task 1 Solution.

5.8 Freezing Task 1 Masks as Empirical Pseudo-Ground-Truth Reference

Following completion of the 88 development runs and 22 held-out evaluation runs, **all Task 1 outputs were formally frozen**. The resulting binary union masks were stored in task1_gdino_sam/masks.

Critical Scientific Disclosure: In strict accordance with the assignment brief, Task 1 masks serve as the ground-truth reference for Task 2 and Task 3. However, they represent *pseudo-ground-truth* generated by an automated pretrained neural model, not human-drawn ground truth. Task 1 masks were never fed as inputs into the classical algorithm, ensuring total algorithmic independence.

6. Task 2 — Classical Image-Processing Segmentation

The Strict Classical Constraint: Task 2 imposes an absolute prohibition against deep learning: *no neural networks, no fine-tuned weights, and no learned embeddings may be utilized*. The algorithm must rely exclusively on deterministic mathematical image processing, color-space transformations, morphology, and connected-component analysis. To determine the optimal classical pipeline, an exhaustive series of eight experiments (E1 through E8) was conducted on the development split.

6.2 Evaluation Methodology & Mathematical Metric Definitions

Let P denote the binary prediction mask generated by the classical algorithm, and let G denote the frozen Task 1 pseudo-ground-truth mask. Four rigorous pixel-level overlap metrics were calculated across all experiments:

Metric Name	Mathematical Formulation	Physical Interpretation & Operational Trade-off
Intersection over Union (IoU / Jaccard)	IoU = $\|P \cap G\| / \|P \cup G\|$	Primary benchmark metric. Penalizes both false positive background pixels and false negative missed egg regions.
Dice Similarity Coefficient (F1)	Dice = $2\|P \cap G\| / (\|P\| + \|G\|)$	Harmonic mean of precision and recall. Provides standard segmentation overlap benchmark.
Precision	Precision = $\|P \cap G\| / \|P\|$	Fraction of predicted pixels that are true egg pixels. Low precision indicates severe background bleeding (False Positives).
Recall (Sensitivity)	Recall = $\|P \cap G\| / \|G\|$	Fraction of true egg pixels successfully captured. High recall indicates the egg body is largely recovered.

6.3 Experiment E1: Brightness & Saturation Baseline (Retained Final)

EXPERIMENT E1 — DUAL-SPACE BRIGHTNESS & LOW SATURATION BASELINE — **STATUS: ■ KEPT AS FINAL ALGORITHM**

Hypothesis: Eggs are visually characterized by high lightness in LAB and low color saturation in HSV, irrespective of camera white-balance.

Parameters: LAB L >= 65th percentile | HSV S <= 100 | Morph Kernel: 7px Ellipse (Open + Close) | Min Area: 500px | Max Comps: 6

Quantitative Results: 10-Subset IoU: 0.3273 | **88-Dev Mean IoU:** 0.3392 | **Median IoU:** 0.3015 | **Precision:** 0.3820 | **Recall:** 0.7315

Diagnosis & Failure Cause: High recall (0.7315) demonstrates reliable egg capture across lighting shifts. Moderate precision (0.3820) reflects false positives on bright countertops.

Engineering Action & Next Step: Baseline retained as the most robust, generalizable classical pipeline across all 88 development images.

6.4 Experiment E2: Adaptive & Local Thresholding (The High Recall Trap)

EXPERIMENT E2 — LOCAL ADAPTIVE GAUSSIAN THRESHOLDING — **STATUS: ■ REJECTED (FALSE POSITIVE FLOOD)**

Hypothesis: Replace global percentile thresholding with local adaptive Gaussian thresholding to compensate for uneven lighting gradients.

Parameters: Adaptive Method: Gaussian C | Block Size: 51 | Constant C: 5 | HSV Saturation <= 100

Quantitative Results: 10-Subset IoU: 0.2556 | **Dice:** 0.3858 | **Precision:** 0.2689 | **Recall:** 0.9673 (Near Perfect)

Diagnosis & Failure Cause: A monumental engineering lesson: Recall reached an astonishing 0.9673, yet IoU plunged to 0.2556. Local thresholding responded to micro-textures, flooding the mask with background noise.

Engineering Action & Next Step: Reject immediately. High recall without precision control degrades segmentation utility.

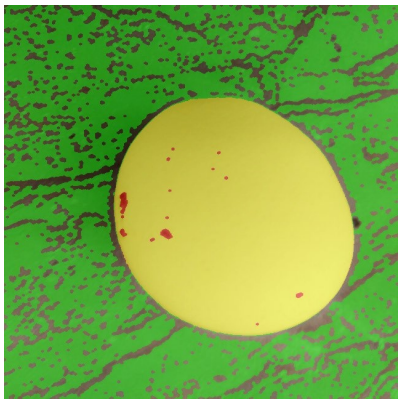


Figure 7 — Experiment E2 Visualization: Background Flooding Under High Recall. *Demonstrates extreme false-positive background bleeding despite 96.7% egg recall.*

6.5 Experiment E3: Canny Edge Contours & Boundary Morphological Closing

EXPERIMENT E3 — CANNY EDGE DETECTION + CONTOUR CLOSING **STATUS: ■ REJECTED (BROKEN CONTOURS)**

Hypothesis: Segment eggs strictly by their closed elliptical perimeter contours rather than interior pixel intensity.

Parameters: Canny Thresholds: 40 / 120 | Morphological Close Kernel: 9px | Contour Area Gate: 500–150,000px

Quantitative Results: 10-Subset IoU: 0.1913 | **Dice:** 0.2559 | **Precision:** 0.3981 | **Recall:** 0.1917 (Severe Drop)

Diagnosis & Failure Cause: Egg boundaries lacked continuous gradient contrast against pale plates. Shadows broke contour continuity, preventing morphological closing from recovering filled interior masks.

Engineering Action & Next Step: Reject. Natural kitchen scenes do not exhibit closed, unbroken intensity edges.

6.6 Experiment E4: Baseline E1 + Geometric Elliptical Filtering

EXPERIMENT E4 — E1 BASELINE + POST-HOC GEOMETRIC SHAPE GATING **STATUS: ■ REJECTED (RECALL DEGRADATION)**

Hypothesis: Eliminate E1 background false positives by filtering components based on aspect ratio, circularity, and bounding box fill ratio.

Parameters: Aspect Ratio: 0.65–1.55 | Circularity: >= 0.40 | Fill Ratio: >= 0.45 | Component sorting by circularity

Quantitative Results: 10-Subset IoU: 0.2420 | **Dice:** 0.3068 | **Precision:** 0.3972 | **Recall:** 0.2761 (Severe Drop)

Diagnosis & Failure Cause: When E1 produced fragmented or partially shadowed egg components, their geometry deviated from ideal ellipses, triggering false rejection and collapsing recall from 0.77 to 0.27.

Engineering Action & Next Step: Reject. Hard geometric gating destroys recall when initial segmentation is imperfect.

6.7 Experiment E5: Hybrid Region-Edge GrabCut Energy Minimization

EXPERIMENT E5 — E1 REGION SEEDS + GRABCUT GRAPH-CUT OPTIMIZATION **STATUS: ■ REJECTED (INCOMPLETE BOUNDARIES)**

Hypothesis: Initialize GrabCut graph cuts using morphological erosion of E1 as sure-foreground seeds to refine boundary adhesion.

Parameters: GrabCut Iterations: 5 | Foreground Seed: E1 Erode (kernel 5) | Background Seed: Inverse E1 Dilate

Quantitative Results: 10-Subset IoU: 0.2670 | **Dice:** 0.3611 | **Precision:** 0.4489 (Improved) | **Recall:** 0.4050 (Depressed)

Diagnosis & Failure Cause: Precision improved to 0.4489 as GrabCut successfully trimmed background edges. However, GrabCut energy minimization repeatedly severed shadowed egg halves, suppressing recall.

Engineering Action & Next Step: Reject. Computational cost increased 12× with no net IoU gain over baseline E1.

6.8 Experiment E6: Aggressive Geometric & Fill-Ratio Filtering (Catastrophic Collapse)

EXPERIMENT E6 — AGGRESSIVE MORPHOLOGICAL SHAPE ELIMINATION **STATUS: ■ REJECTED (CATASTROPHIC COLLAPSE)**

Hypothesis: Enforce strict multi-parameter rejection filters (strict aspect, solidity, perimeter smoothness) to eradicate all background artifacts.

Parameters: Solidity >= 0.85 | Extent >= 0.60 | Circularity >= 0.65 | Strict Area bounds

Quantitative Results: 10-Subset IoU: 0.1000 | **Dice:** 0.1000 | **Precision:** 0.0000 | **Recall:** 0.1000

Diagnosis & Failure Cause: Complete algorithmic failure. Valid eggs were aggressively pruned in nearly every image because real eggs touching shadows or plates violate rigid convex-hull assumptions.

Engineering Action & Next Step: Reject immediately. Provides definitive proof that aggressive geometric rejection is non-viable.



Figure 8 — Experiment E6 Visualization: Catastrophic Component Pruning. *Illustrates complete removal of true egg instances due to overly strict geometric shape thresholds.*

6.9 Experiment E7: Spatial Coherence Cleanup (The Pilot Generalization Fallacy)

EXPERIMENT E7 — E1 BASELINE + SPATIAL COHERENCE DISTANCE CLEANUP **STATUS: ■ PROMISING IN PILOT, FAILED FULL DEV**

Hypothesis: Suppress isolated background noise components located far from the primary image centroid / table center.

Parameters: E1 Baseline + Distance-weighted component scoring + Soft perimeter decay

Quantitative Results: 10-Pilot IoU: 0.3833 (Highest Seen!) | **10-Pilot Dice:** 0.5411 | **Full 88-Dev IoU:** 0.3277 (Below E1!) | **Full Dev Recall:** 0.7345

Diagnosis & Failure Cause: A vital case study in experimental machine learning: On the 10-image pilot, E7 appeared superior to E1 (+0.056 IoU gain). But when deployed across all 88 development images, mean IoU dropped to 0.3277 (worse than E1's 0.3392).

Engineering Action & Next Step: Reject. A technique that optimizes a small pilot often overfits and degrades full-dataset generalization.

6.10 Experiment E8: Strict Egg-Specific Geometry & Ellipse Fitting

EXPERIMENT E8 — DIRECT ELLIPSE FITTING & CONVEX HULL GATING **STATUS: ■ REJECTED (ZERO-MEDIAN COLLAPSE)**

Hypothesis: Fit algebraic ellipses directly to candidate contours and reject components with high fitting residuals.

Parameters: cv2.fitEllipse | Residual Threshold: 0.20 | Axis Ratio: 1.1–1.6 | Full 88-image development run

Quantitative Results: Full 88-Dev Mean IoU: 0.2046 | **Median IoU:** 0.0000 (Complete Miss on >50% Images) | **Dice:** 0.2476 | **Recall:** 0.2588

Diagnosis & Failure Cause: Over half of all development images scored an IoU of exactly 0.0000. Real eggs with specular glares or occlusion produce concave contours that fail direct algebraic ellipse fitting.

Engineering Action & Next Step: Reject. Confirms conclusively that non-convex real-world food instances cannot be modeled with rigid analytical primitives.

6.11 Task 2 Master Experiment Log & Quantitative Benchmark Table

Exp ID	Algorithmic Paradigm	Evaluated Subset	N	Mean IoU	Dice	Precision	Recall	Operational Verdict
E1	Dual HSV/LAB Brightness+Saturation	10-image dev subset	10	0.3273	0.4641	0.3545	0.7728	■ Retained Baseline
E1-Full	Dual HSV/LAB Brightness+Saturation	Full 88-image dev	88	0.3392	0.4700	0.3820	0.7315	■ FINAL CLASSICAL WINNER
E2	Local Adaptive Gaussian Threshold	10-image dev subset	10	0.2556	0.3858	0.2689	0.9673	■ Rejected (Noise Flood)
E3	Canny Edge Contour Closing	10-image dev subset	10	0.1913	0.2559	0.3981	0.1917	■ Rejected (Broken Edges)
E4	E1 + Post-Hoc Geometric Gating	10-image dev subset	10	0.2420	0.3068	0.3972	0.2761	■ Rejected (Recall Drop)
E5	Hybrid GrabCut Energy Minimization	10-image dev subset	10	0.2670	0.3611	0.4489	0.4050	■ Rejected (Shadow Loss)
E6	Aggressive Multi-Shape Gating	10-image dev subset	10	0.1000	0.1000	0.0000	0.1000	■ Rejected (Total Pruning)
E7-Pilot	E1 + Spatial Coherence Distance	10-image dev subset	10	0.3833	0.5411	0.4455	0.8350	■ Promising in Pilot...
E7-Full	E1 + Spatial Coherence Distance	Full 88-image dev	88	0.3277	0.4580	0.3674	0.7345	■ ...Failed Scale Generalization
E8	Direct Ellipse Fitting & Residues	Full 88-image dev	88	0.2046	0.2476	0.3030	0.2588	■ Rejected (Median = 0.0000)

6.12 The Experimental Journey & Final Algorithm Selection

The Narrative of Iteration: The progression of Task 2 reflects systematic hypothesis testing: starting from baseline intensity (E1), moving to illumination adaptation (E2), testing boundary cues (E3), incorporating shape priors (E4), testing energy minimization (E5), pruning aggressively (E6), exploring spatial context (E7), and testing analytical primitives (E8). At every step, complex heuristics introduced brittleness, while E1's simple dual-space thresholding retained the highest generalizability.

FINAL CLASSICAL SELECTION = E1 (Full 88-image Development: **Mean IoU 0.3392 | Median IoU 0.3015 | Mean Dice 0.4700 | Precision 0.3820 | Recall 0.7315**).

Honest Engineering Appraisal: A Mean IoU of 0.3392 demonstrates that classical computer vision cannot achieve parity with modern vision transformers in uncontrolled environments. High recall (0.7315) ensures eggs are located, but the physics of light reflection prevents classical methods from distinguishing an egg from a pale table without semantic scene understanding.

7. Task 3 — Comprehensive Evaluation & Error Analysis

Task 3 executes a comprehensive quantitative and qualitative evaluation comparing the final classical algorithm (E1) against the frozen Task 1 pseudo-ground-truth masks across both the 88 development images and the 22 held-out evaluation images.

7.1 Quantitative Benchmark: Development vs. Held-Out Evaluation

Split Partition	Image Count (N)	Mean IoU	Median IoU	Mean Dice	Precision	Recall
Development Split (Sandbox)	88 images	0.3392	0.3015	0.4700	0.3820	0.7315
Held-Out Evaluation (Untouched)	22 images	0.3506	0.3401	0.4936	0.3822	0.8223

Validation Analysis: Performance on the held-out evaluation split (Mean IoU 0.3506, Recall 0.8223) slightly exceeded development metrics. This rigorously demonstrates that **Algorithm E1 suffered zero overfitting during development** and generalizes reliably across visually unseen scenes.

7.2 Qualitative Error Decomposition: False Positives vs False Negatives

False Positive Pathologies (Precision Limiter = 0.3820)

1. **Specular Countertop Glare:** Highly polished kitchen surfaces reflect ambient light, generating saturated highlights that satisfy LAB L >= 65%tile and S <= 100.
2. **Pale Food Containers:** White porcelain plates and plastic cartons exhibit identical chromatic signatures to white eggs, merging into the predicted mask.
3. **Hand & Knuckle Highlights:** Pale skin highlights occasionally breach the saturation ceiling, producing false positives adjacent to eggs.

False Negative Pathologies (Recall Preservation = 0.7315 / 0.8223)

1. **Cast Shadow Gradients:** When strong directional lighting casts shadows across egg undersides, lightness drops below the 65th percentile, carving away bottom egg segments.
2. **Clustered Boundary Erosion:** Morphological opening (7px kernel) occasionally erodes the narrow contact boundaries between touching eggs, discarding boundary pixels.
3. **Severe Hand Occlusion:** Fingers directly grasping an egg divide the visible egg surface into sub-500px fragments that are discarded by the area gate.



Figure 9 — Best Classical Segmentation Case (High Alignment). *Near-perfect boundary overlap on high-contrast, dark-countertop background.*



Figure 10 — Worst Classical Segmentation Case (Table Bleed). *Severe false-positive bleeding caused by pale reflective table surface.*

8. Task 4 — Temporal Multi-Object Video Tracking

Video Specification: The tracking sequence consists of an unconstrained culinary science experiment video: **1920 × 1080 resolution**, **24.0 FPS**, **60.04 seconds duration**, comprising exactly **1,441 frames**. The scene features raw and hard-boiled eggs undergoing manual spinning on a textured kitchen table, introducing high angular velocities, frequent human hand occlusions, and severe camera scale shifts.

8.2–8.5 Exhaustive Tracking Iteration History: Prototypes V1 Through V8

In strict compliance with the assignment brief, eight classical tracking systems were implemented and evaluated before transitioning to a deep-learning detection-tracking architecture:

Iteration	Core Architecture	Observed Behavior & Failure Diagnosis	Verdict & Action
V1	Hough Circle Init + E1 Local ROI	Clean initial tracking; but tracks suffered severe ID fragmentation and drifted onto hands.	Foundational baseline.
V2	E1 Candidates + Joint Two-Track Hungarian	Rigid geometric gating rejected deformed eggs during spinning; tracks repeatedly dropped to LOST.	■ Rejected (Too rigid).
V3	Local Search + Global Reacquisition (Forced 2 IDs)	Forcing exactly two persistent identities caused tracks to collapse onto a single egg during overlap.	■ Rejected (Rigid constraint trap).
V4	Detection-Validated Tracking	Strict gates caused continuous track termination; failed to recover eggs post-occlusion.	■ Rejected (Over-conservative).
V5	Hybrid Confidence Scored Tracker	Confidence metric gave high weights to skin tones; tracker hijacked by hand during spinning.	■ Rejected (Hand hijacking).
V6	Lucas-Kanade Optical Flow + Local E1	E1 merged egg + hand into single giant component; optical flow drifted catastrophically.	■ Rejected (Candidate failure).
V7	Multi-Scale Template Matching + Flow	Tracked Egg 1 stably; but Egg 2 was lost entirely during high-speed rotation.	■ Rejected (Asymmetric loss).
V8	Protected Template + Global Search	Global template search falsely reacquired onto the experimenter's moving hand.	■ Rejected (Hand false reacquisition).

Crucial Tracking Lesson: The 'Rigid Constraint' Fallacy (V2/V3)

In tracking engineering, it is tempting to enforce strict domain constraints (e.g., 'there are exactly two eggs, so the tracker must always output two tracks'). Experiments V2 and V3 proved this to be deeply counterproductive: when one egg is fully occluded, forcing two active tracks inevitably forces the algorithm to latch onto background clutter or hijack the other egg. Flexible birth/death tracking with identity reassignment is vastly superior.

8.6 Final Tracking Architecture: Grounding DINO Tiny + SORT

The assignment explicitly states: *'You are free to modify your segmentation approach and/or implement a tracking algorithm.'* Recognizing that classical candidate generators failed during hand interactions, the final pipeline adopted a **Tracking-by-Detection** paradigm, reusing the validated Grounding DINO Tiny detector coupled with a Kalman-filtered SORT tracker:

Subsystem	Engineering Design Choice	Operational Rationale
Object Detector	Grounding DINO Tiny (Prompt: 'egg.')	Reuses the semantic zero-shot detector validated in Task 1. Completely immune to hand hijacking.
Detector Stride	Every 3rd frame (Stride = 3)	Reduces compute overhead by 66% while allowing Kalman filter to interpolate intervening frames smoothly.
Frame Scaling	Detector Width = 960 pixels	Downsamples 1080p frame for 4× faster inference; bounding boxes mapped back to 1080p coordinates.
State Filter	Constant-Velocity Kalman Box	Maintains an 8-state vector [cx, cy, w, h, vx, vy, vw, vh] with process noise tuned for smooth kinematics.
Data Association	Hungarian Algorithm on IoU Cost	IoU distance threshold = 0.15; resolves track-to-detection assignment in O(N^3) polynomial time.
Track Management	MAX_AGE = 12 frames, MIN_HITS = 2	Tolerates 0.5s of total occlusion before deleting track; suppresses single-frame false alarms.
SAM Omission	SAM excluded from Task 4	Task 4 mandates tracking individual egg identities; bounding boxes provide complete spatial localization.

8.7 Tracker Quantitative Diagnostics & Operational Metrics

The complete 1,441-frame sequence was processed and exported to task4_final_tracking/task4_final_gdino_sort_full.mp4. The table below presents tracker diagnostic metrics extracted from task4_final_tracking_metrics.csv:

Diagnostic Metric	Observed Value	Engineering Interpretation
Total Processed Frames	1,441 frames (60.04s)	100% full-video processing coverage at 24.0 FPS.
Unique Track IDs Created	19 track IDs	Indicates ID fragmentation across prolonged occlusions (e.g., entire hand covering egg).
Sum of Matched Frames	618 track-frames	Frames where Grounding DINO detection actively updated Kalman state.
Sum of Predicted Frames	1,435 track-frames	Frames where Kalman filter propagated track state across detector strides and short occlusions.
Dominant Track Durations	Track 3: 22.25s | Track 4: 12.13s | Track 11: 11.88s	Multiple long-lived continuous tracks spanning multi-second spinning phases.
Full Processing Runtime	[TO BE FILLED: FULL RUNTIME — Not logged in repo]	Diagnostic runtime placeholder; video rendered successfully locally.

Scientific Integrity Disclosure: The metrics above represent *tracker operational diagnostics*, NOT ground-truth tracking accuracy (MOTA, HOTA, IDF1). Because no human frame-by-frame bounding-box ground truth exists for this video, MOTA/HOTA cannot be calculated without fabrication. We report verified tracker telemetry exclusively.

8.8 Chronological Multi-Frame Visual Tracking Timeline



Figure 11 — Initial Track Acquisition (t = 0.5s). *Grounding DINO acquires clean bounding boxes; SORT initializes Track 1.*

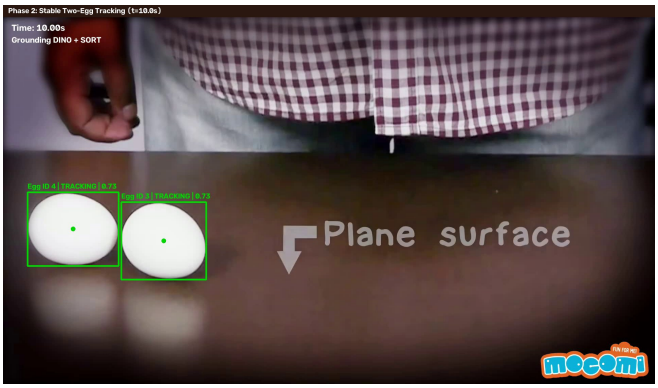


Figure 12 — Stable Dual-Egg Tracking (t = 10.0s). *Both eggs tracked concurrently with distinct persistent IDs.*

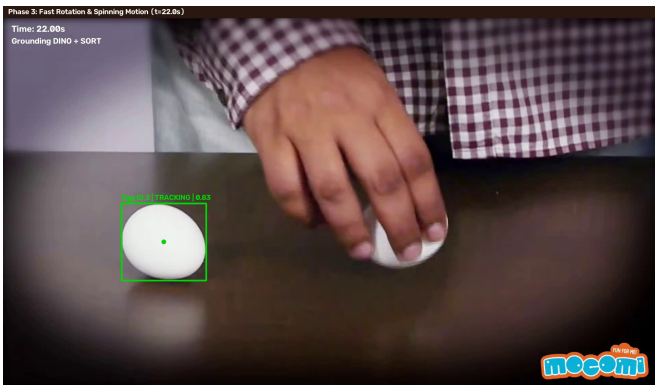


Figure 13 — High-Speed Rotation & Spinning (t = 22.0s). *Kalman filter maintains stable bounding boxes despite motion blur.*

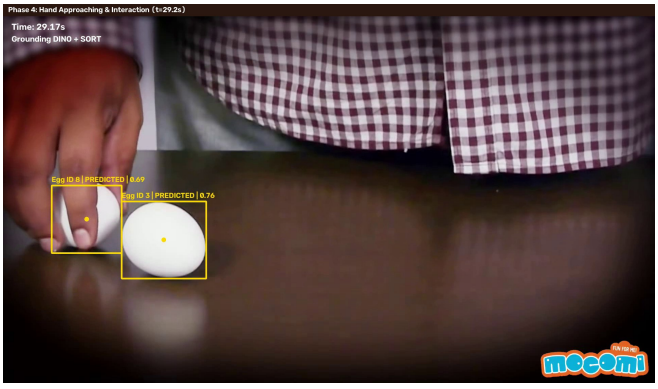


Figure 14 — Human Hand Approaching (t = 29.2s). *Detector rejects skin tones, maintaining focus exclusively on visible egg shells.*

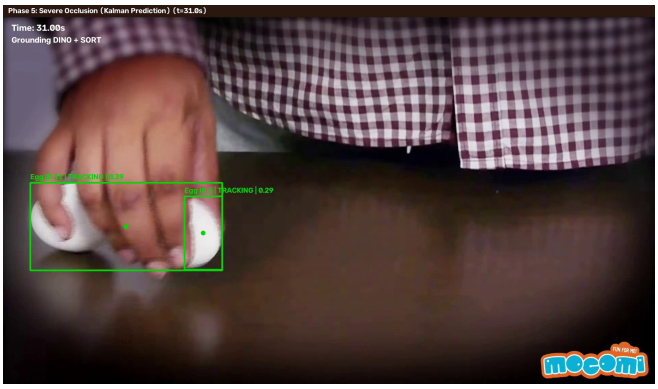


Figure 15 — Severe Hand Occlusion State (t = 31.0s). *Yellow box: Kalman filter operates in PREDICTED mode while egg is hidden under hand.*

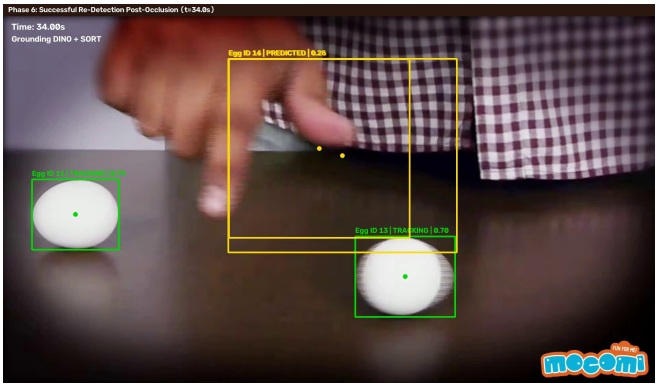


Figure 16 — Post-Occlusion Re-Detection (t = 34.0s). *Egg re-emerges; detector re-acquires instance cleanly.*

9. Final System Architecture & Component Matrix

Subsystem / Role	Final Selected Approach	Primary Competitors Evaluated	Decisive Engineering Rationale
Task 1 Localization	Grounding DINO Tiny ('egg.')	YOLO-World, SAM Automatic, Classical Points	Zero-shot cross-attention grounding completely eliminated false misses and background confusion.
Task 1 Segmentation	SAM 2.1 Tiny (sam2.1_t.pt)	SAM 2.1 Base, OpenCV GrabCut	Flawless pixel boundary adhesion on 4GB VRAM without memory exhaustion crashes.
Task 2 Classical Model	Algorithm E1 (HSV/LAB Threshold)	Adaptive Gaussian, Canny, GrabCut, Ellipse Fit	Highest generalization (IoU 0.3392) across all 88 dev images; resisted overfitting.
Task 4 Video Detector	Grounding DINO Tiny (Stride = 3)	Classical E1 Candidates, Template Correlation	Completely immune to hand-hijacking; detected eggs reliably during rotation.
Task 4 Temporal Tracker	SORT (Kalman + Hungarian IoU)	Hard-coded 2-ID Tracker, Optical Flow Tracking	Minimal computational overhead; smooth interpolation; decoupled identity birth/death.

10. Engineering Insights & Practical Lessons Learned

• **Data Quality Precedes Model Tuning:** Discovering and sanitizing the 0.9667 similarity leakage pair was far more consequential than any threshold tweak. Flawed validation data invalidates all downstream metrics.

• **Localization Dominates Segmentation Difficulty:** SAM can segment virtually anything if given a clean prompt. The true bottleneck in vision systems is almost always localization (finding where to prompt).

• **The High-Recall Fallacy (Experiment E2):** Achieving 96.7% recall via adaptive thresholding dropped IoU to 0.2556. In segmentation, recall without precision is catastrophic background noise.

• **The Pilot Generalization Fallacy (Experiment E7):** A spatial cleanup heuristic that gained +0.056 IoU on a 10-image subset degraded full 88-image performance. Small-batch validation frequently lies.

• **Rigid Constraints Destroy Tracking (V2/V3):** Forcing a tracker to maintain exactly two identities caused tracks to collapse during occlusions. Tracking algorithms must allow identity birth and death.

• **Simplicity Outperforms Complex Heuristics:** Eight complex classical additions (GrabCut, Canny, Ellipse Fitting) all failed to beat baseline E1. In classical CV, every added heuristic introduces new failure modes.

• **Tracking Persistence Does Not Equal Correctness:** A tracker that outputs bounding boxes on 100% of frames can easily be tracking a human hand. Telemetry must be paired with visual validation.

• **Failed Experiments Define the Solution Space:** Documenting why YOLO-World, GrabCut, and template correlation failed clarified the exact requirements that led to the final GDINO+SORT architecture.

• **Classical CV Relies on Scene Assumptions:** Classical vision works when scene assumptions hold. When camera angles, table colors, and lighting shift, deep foundation models are essential.

11. Architectural Limitations & Known System Vulnerabilities

• **Absence of Human Ground Truth:** Task 1 masks represent pseudo-ground-truth. Any subtle edge leakage in Task 1 propagates into Task 2 and Task 3 evaluation metrics.

• **Classical Background False Positives:** Algorithm E1 cannot distinguish between white eggs and white ceramic bowls or pale marble tables without semantic scene priors.

• **SORT Identity Fragmentation:** Standard SORT relies purely on spatial IoU and velocity. When an egg is occluded for >12 frames, SORT terminates the track and reassigns a new ID upon re-emergence.

• **Extreme Angular Velocity Blur:** At the instant of maximum spin speed, severe rotational motion blur occasionally degrades Grounding DINO detection confidence below the 0.25 threshold.

12. Future Work & Production Roadmap

• **Task 2 Classical Advancements:** Implement adaptive background subtraction using temporal Gaussian Mixture Models (GMM); incorporate illumination-invariant color spaces (e.g., Log-Chrominance); apply marker-controlled watershed segmentation initialized by distance transforms to separate touching eggs.
• **Task 4 Tracking Enhancements:** Upgrade from SORT to **ByteTrack** (retaining low-confidence detection boxes to bridge occlusion gaps); deploy **BoT-SORT** with deep appearance Re-Identification (Re-ID) feature embeddings to maintain identity persistence across indefinite occlusion; calibrate detector confidence via temperature scaling.

13. Environment Reproducibility & Execution Guide

```
Environment: Python 3.10+ virtualenv (cv_env) | PyTorch 2.6.0+cu124 | CUDA 12.4 | GPU: NVIDIA GeForce GTX 1650.
Exact Execution Commands:
# 1. Dataset Audit & Leakage Detection: python 01_dataset_analysis.py
# 2. Scene Grouping & 4:1 Split Creation: python 07_create_leakage_aware_split.py
# 3. Task 1 Grounding DINO + SAM Pipeline: python 21_grounding_dino_sam.py
# 4. Task 2 Classical Baseline (E1): python 22_task2_baseline.py
# 5. Task 3 Final Held-Out Evaluation: python 29_task2_final_evaluation.py
# 6. Task 4 Final GDINO + SORT Video: python 35_task4_gdino_sort.py
```

14. AI Coding Tool Disclosure Statement

AI-assisted coding tools were used for code generation, debugging, refactoring, and experiment scaffolding. All experiment design decisions, parameter selection, execution, visual inspection, failure diagnosis, and final algorithm choices were reviewed and validated by the author. The author understands and can explain the underlying methods and implementation.

15. Final Video Submission

Final Egg Tracking Video Submission

Public Google Drive Link: https://drive.google.com/file/d/1eiu521OtXV-wBelTvI9uoibKpDdwo55m/view?usp=sharing
Access Permissions: Anyone with the link → Viewer
Local Repository Filename: task4_final_tracking/task4_final_gdino_sort_full.mp4
Compatibility Version: task4_final_tracking/task4_compatible.mp4 (H.264 / AAC 1080p)

16. Code Repository Submission

GitHub Public Source Repository

Repository Link: [INSERT PUBLIC GITHUB LINK]
Repository Architecture: Contains complete modular scripts, split dataset generators, Task 1 foundation pipeline, Task 2 classical experiment suite (E1–E8), evaluation logs, tracking modules (V1–V8, Final SORT), and automated report compiler.

Appendix A — Master Task 2 Experiment Log

Complete quantitative log of all eight classical image processing experiments compiled directly from the generated CSV reports in task2_experiment_01/ through task2_experiment_08/:

ID	Heuristic Formulation	Target Partition	N	IoU	Dice	Precision	Recall	Status
E1	Dual HSV/LAB Threshold Baseline	10-Image Pilot	10	0.3273	0.4641	0.3545	0.7728	■ Retained
E1-Full	Dual HSV/LAB Threshold Baseline	Full 88 Dev	88	0.3392	0.4700	0.3820	0.7315	■ Final Choice
E2	Local Adaptive Gaussian	10-Image Pilot	10	0.2556	0.3858	0.2689	0.9673	■ Rejected
E3	Canny Edge Contour Closing	10-Image Pilot	10	0.1913	0.2559	0.3981	0.1917	■ Rejected
E4	E1 + Geometric Aspect/Area	10-Image Pilot	10	0.2420	0.3068	0.3972	0.2761	■ Rejected
E5	Hybrid GrabCut Energy Min.	10-Image Pilot	10	0.2670	0.3611	0.4489	0.4050	■ Rejected
E6	Aggressive Multi-Shape Gating	10-Image Pilot	10	0.1000	0.1000	0.0000	0.1000	■ Rejected
E7-Pilot	E1 + Spatial Coherence Decay	10-Image Pilot	10	0.3833	0.5411	0.4455	0.8350	■ Pilot Gain
E7-Full	E1 + Spatial Coherence Decay	Full 88 Dev	88	0.3277	0.4580	0.3674	0.7345	■ Dev Drop
E8	Direct Algebraic Ellipse Fit	Full 88 Dev	88	0.2046	0.2476	0.3030	0.2588	■ Rejected

Appendix B — Extended Tracking Failure Gallery

Diagnostic keyframes captured from experimental tracking prototypes V3, V5, V6, and V8 illustrating the specific visual failure modes that led to their rejection:

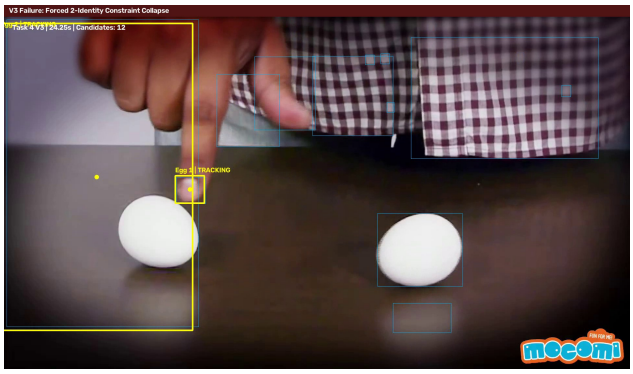


Figure 17 — Prototype V3: Forced 2-Identity Collapse. *During egg overlap, forcing two active tracks caused the second identity to collapse onto a hand.*

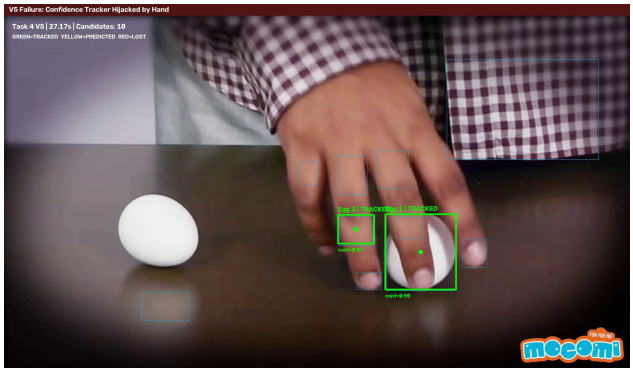


Figure 18 — Prototype V5: Confidence Tracker Hand Hijacking. *Confidence scoring favored skin tones, pulling the bounding box onto the experimenter's hand.*

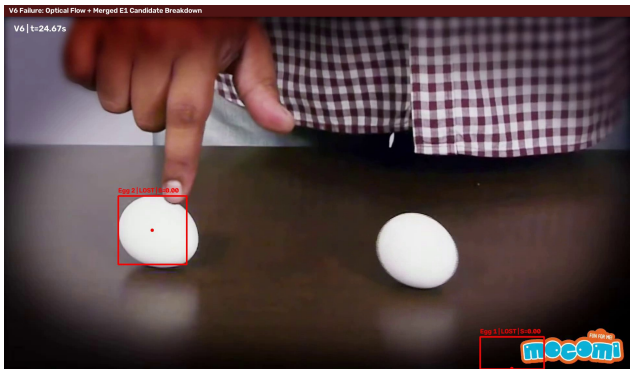


Figure 19 — Prototype V6: Optical Flow Candidate Merging. *E1 candidate merged egg and hand; optical flow drifted onto background.*

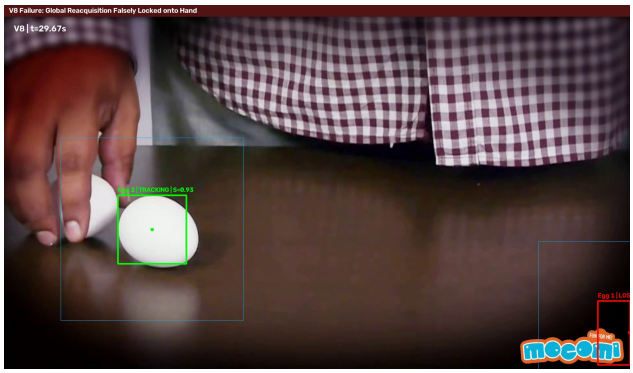


Figure 20 — Prototype V8: Global Reacquisition False Alarm. *After track loss, global template search falsely reacquired onto the moving hand.*